

D236

Pathophysiology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A 12-year-old who plays travel soccer has weight loss, extreme thirst, frequent urination, and ketones in her urine. Her older brother has asthma. What is the most likely underlying mechanism?

- A. Autoimmune destruction of pancreatic beta cells, causing absolute insulin deficiency
- B. A side effect of her brother's asthma medication
- C. Insulin resistance in muscle and fat tissue
- D. Excess glucagon with normal insulin

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which mechanism best describes early type 2 diabetes?

- A. Insulin resistance with relative insulin deficiency
- B. No insulin is produced at all
- C. Too much insulin causing low blood glucose
- D. Autoimmune destruction of beta cells

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A carpenter who recently bought a new truck cuts his hand. Within hours the area is red, warm, swollen, and painful. He had a tetanus booster last year. What causes the swelling?

- A. A reaction to the tetanus booster
- B. Increased vascular permeability lets fluid and proteins leak into the tissue.
- C. Vasoconstriction traps blood in the area.
- D. Bacterial infection is always present when swelling occurs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A teacher at an elementary school with a new playground is stung by a bee and within minutes develops hives, wheezing, and low blood pressure. She was stung once before without problems. Which hypersensitivity type is this?

- A. Type IV, delayed T-cell
- B. Not an allergy, since her first sting caused no problem
- C. Type I, IgE-mediated
- D. Type III, immune complex

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A 68-year-old retired mail carrier with a history of heart attack has crackles in both lungs, shortness of breath when lying flat, and pink frothy sputum. He enjoys gardening. Which condition is most likely?

- A. Bacterial pneumonia
- B. Right-sided heart failure
- C. Left-sided heart failure
- D. Right-sided failure, since heart failure always starts on the right

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which findings point to right-sided heart failure?

- A. Wheezing from airway spasm
- B. Jugular vein distention, leg edema, and an enlarged liver
- C. High blood pressure only
- D. Crackles and shortness of breath lying flat

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A patient with COPD who used to work in a textile mill has these blood gases: pH 7.30, PaCO₂ 58 mm Hg, HCO₃ 26 mEq/L. He lives with his wife and two cats. What is the interpretation?

- A. Fully compensated respiratory acidosis
- B. Metabolic acidosis
- C. Respiratory alkalosis
- D. Uncompensated respiratory acidosis

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Blood gases show pH 7.50, PaCO₂ 30 mm Hg, HCO₃ 24 mEq/L. Which condition fits?

- A.** Respiratory alkalosis, as with hyperventilation
- B.** Respiratory acidosis
- C.** Metabolic alkalosis
- D.** Normal, because HCO₃ is normal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A 50-year-old librarian who recently moved to a colder city reports fatigue, weight gain, feeling cold, constipation, and dry skin. Her TSH is high and her free T₄ is low. What is the most likely diagnosis?

- A.** A normal response to a colder climate
- B.** Hyperthyroidism
- C.** Secondary hypothyroidism from pituitary failure
- D.** Primary hypothyroidism

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which lab pattern fits primary hyperthyroidism, such as Graves disease?

- A.** A normal TSH always rules out thyroid disease
- B.** High TSH, low free T₄
- C.** High TSH, high free T₄
- D.** Low TSH, high free T₄

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A 72-year-old with atrial fibrillation who volunteers at a museum develops sudden right-sided weakness and slurred speech. A CT scan shows no bleeding. His wife drove him in. What is the most likely type and cause?

- A.** Hemorrhagic stroke from a ruptured vessel
- B.** A migraine with aura
- C.** Ischemic stroke from a clot that formed in the heart
- D.** A transient ischemic attack, since onset was sudden

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A 28-year-old with heavy menstrual periods has fatigue and low hemoglobin with small, pale red blood cells (low MCV). She works as a graphic designer. What is the most likely diagnosis?

- A.** Aplastic anemia
- B.** Iron deficiency anemia
- C.** Any anemia, since all anemias cause large cells
- D.** Vitamin B12 deficiency anemia

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A 12-year-old ~~who plays travel soccer~~ has weight loss, extreme thirst, frequent urination, and ketones in her urine. ~~Her older brother has asthma.~~ What is the most likely underlying mechanism?

Target word: mechanism **Decides:** Child, weight loss, and ketones point to absolute insulin deficiency.

A	Correct	Classic type 1 diabetes.
B	Camouflage	Her brother's condition does not affect her.
C	Confusion set	That is the main mechanism of type 2.
D	Adjacent category	Not the mechanism of this presentation.

2. Answer A Concept 1 • transfer

Which mechanism best describes early type 2 diabetes?

Target word: type 2 mechanism **Decides:** Cells respond poorly to insulin.

A	Correct	The pancreas still makes insulin, but tissues respond poorly.
B	Absolute language	Early type 2 often has normal or high insulin levels.
C	Inverted operation	Type 2 causes high glucose.
D	Confusion set	That is type 1.

3. Answer B Concept 2

A carpenter ~~who recently bought a new truck~~ cuts his hand. Within hours the area is red, warm, swollen, and painful. ~~He had a tetanus booster last year.~~ What causes the swelling?

Target word: swelling mechanism **Decides:** Fluid leaks from more permeable vessels.

A	Camouflage	A year-old vaccine does not explain local swelling.
B	Correct	Leaky vessels produce edema.
C	Inverted operation	Inflammation causes vasodilation.
D	Absolute language	Inflammation occurs with or without infection.

4. Answer C Concept 3

A teacher ~~at an elementary school with a new playground~~ is stung by a bee and within minutes develops hives, wheezing, and low blood pressure. She was stung once before without problems. Which hypersensitivity type is this?

Target word: hypersensitivity type **Decides:** Immediate systemic reaction after prior sensitization.

A	Confusion set	Type IV takes 24–72 hours.
B	Inverted operation	The first exposure sensitizes; the reaction comes later.
C	Correct	Immediate anaphylaxis after a sensitizing first exposure.
D	Adjacent category	Immune complexes cause delayed tissue damage.

5. Answer C Concept 4

A 68-year-old ~~retired mail carrier~~ with a history of heart attack has crackles in both lungs, shortness of breath when lying flat, and pink frothy sputum. ~~He enjoys gardening.~~ Which condition is most likely?

Target word: side of failure **Decides:** Lung congestion signs.

A	Adjacent category	Does not explain orthopnea and frothy sputum after MI.
B	Confusion set	Right failure causes systemic congestion.
C	Correct	Backup into the lungs causes crackles and orthopnea.
D	Absolute language	Left failure is the most common start.

6. Answer B Concept 4 • transfer

Which findings point to right-sided heart failure?

Target word: right-sided signs **Decides:** Systemic venous congestion.

A	Adjacent category	A lung airway problem.
B	Correct	Blood backs up into the body.
C	Dropped element	A risk factor, not the congestion signs.
D	Inverted operation	Those are left-sided signs.

7. Answer D Concept 5

A patient with COPD ~~who used to work in a textile mill~~ has these blood gases: pH 7.30, PaCO₂ 58 mm Hg, HCO₃ 26 mEq/L. ~~He lives with his wife and two cats.~~ What is the interpretation?

Target word: ABG interpretation **Decides:** Low pH, high CO₂, normal HCO₃.

A	Adjacent category	Full compensation would bring pH back into range.
B	Confusion set	HCO ₃ would be low.
C	Inverted operation	pH is low, not high.
D	Correct	CO ₂ retention lowers pH; bicarbonate is still normal.

8. Answer A Concept 5 • transfer

Blood gases show pH 7.50, PaCO₂ 30 mm Hg, HCO₃ 24 mEq/L. Which condition fits?

Target word: ABG interpretation **Decides:** High pH, low CO₂, normal HCO₃.

A	Correct	Blowing off CO ₂ raises pH.
B	Inverted operation	pH is high.
C	Confusion set	HCO ₃ would be high.
D	Dropped element	Ignores pH and PaCO ₂ .

9. Answer D Concept 6

A 50-year-old librarian ~~who recently moved to a colder city~~ reports fatigue, weight gain, feeling cold, constipation, and dry skin. Her TSH is high and her free T₄ is low. What is the most likely diagnosis?

Target word: diagnosis **Decides:** High TSH with low T₄.

A	Camouflage	The labs show thyroid disease.
B	Inverted operation	Would show low TSH and high T ₄ .
C	Adjacent category	TSH would be low.
D	Correct	The pituitary raises TSH because the thyroid underproduces.

10. Answer D Concept 6 • transfer

Which lab pattern fits primary hyperthyroidism, such as Graves disease?

Target word: lab pattern **Decides:** Overactive gland suppresses TSH.

A	Absolute language	Not in all cases.
B	Inverted operation	That is primary hypothyroidism.
C	Adjacent category	Suggests a TSH-secreting pituitary problem.
D	Correct	Excess hormone feeds back to suppress TSH.

11. Answer C Concept 7

A 72-year-old with atrial fibrillation ~~who volunteers at a museum~~ develops sudden right-sided weakness and slurred speech. A CT scan shows no bleeding. ~~His wife drove him in.~~ What is the most likely type and cause?

Target word: stroke type and mechanism **Decides:** Atrial fibrillation and no bleed on CT.

A	Inverted operation	CT shows no bleeding.
B	Confusion set	Does not fit AF plus lasting deficits.
C	Correct	AF promotes emboli; CT shows no hemorrhage.
D	Adjacent category	Sudden onset fits both; TIA symptoms resolve.

12. Answer B Concept 8

A 28-year-old with heavy menstrual periods has fatigue and low hemoglobin with small, pale red blood cells (low MCV). ~~She works as a graphic designer.~~ What is the most likely diagnosis?

Target word: anemia type **Decides:** Low MCV plus chronic blood loss.

A	Adjacent category	Marrow failure lowers all cell lines.
B	Correct	Blood loss depletes iron, producing small, pale cells.
C	Absolute language	Cell size differs by cause.
D	Confusion set	B12 deficiency causes large cells.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Type 1 vs. type 2 diabetes	Type 1 = autoimmune destruction of beta cells, absolute insulin deficiency; type 2 = insulin resistance with relative deficiency.	Calling any diabetes in a young person insulin resistance, or saying type 2 produces no insulin.	Age, ketones, and weight change vs. gradual onset.
2	Acute inflammation	Vasodilation causes redness and heat; increased vascular permeability lets fluid leak, causing swelling.	Attributing swelling to vasoconstriction or assuming infection.	Redness, heat, swelling, pain after injury.
3	Hypersensitivity types	Type I = IgE, immediate (anaphylaxis); II = antibody against cells; III = immune complexes; IV = delayed T-cell.	Thinking a first uneventful exposure rules out allergy.	Minutes after exposure, hives, wheezing, low BP.
4	Left vs. right heart failure	Left failure backs up into the lungs (crackles, orthopnea); right failure backs up into the body (JVD, edema, enlarged liver).	Swapping the signs.	Lung signs vs. systemic congestion.
5	Respiratory acid-base imbalance	Low pH with high PaCO ₂ = respiratory acidosis; high pH with low PaCO ₂ = respiratory alkalosis; normal HCO ₃ = uncompensated.	Calling it metabolic, or calling it compensated when pH is abnormal.	pH, PaCO ₂ , and HCO ₃ values.
6	Thyroid disorders	Primary hypothyroidism: high TSH, low T ₄ , slowed metabolism; primary hyperthyroidism: low TSH, high T ₄ .	Reading high TSH as overactive thyroid.	TSH and T ₄ moving in opposite directions.
7	Ischemic vs. hemorrhagic stroke	Ischemic stroke is a blocked vessel (often an embolus from atrial fibrillation); CT without bleeding supports ischemia.	Assuming sudden onset means hemorrhage.	Atrial fibrillation plus CT result.
8	Anemia by cell size	Iron deficiency causes small, pale cells (low MCV), often from blood loss; B12 or folate deficiency causes large cells.	Mixing up microcytic and macrocytic causes.	MCV and a source of blood loss.

Course scope (public description): Pathophysiology is a course designed for nursing students, providing an overview of the pathology and treatment of diseases in the human body, tissues, glands and membranes, the integumentary system, the sensory system, skeletal and muscular systems, the digestive system, blood, vessels and circulation, lymphatic system, immunity and disease, heart and respiratory system, nervous, urinary and endocrine systems, and male and female reproductive systems. Prerequisites include all prior courses in this programmatic sequence.